

Qatar-2b

R-0966

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_170323 · created 2026-07-29T16:50:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457835.85851 +/- 0.00042 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.226 +/- 0.067
Transit depth (Rp/Rs)^2	5.1 +/- 3.0 [%]
Orbital Inclination (inc)	89.4 +/- 1.75
Airmass coefficient 1 (a1)	0.09 +/- 0.057
Airmass coefficient 2 (a2)	1.52 +/- 0.58
Scatter in the residuals of the lightcurve fit is	72.56 %
Best Comparison Star	None
Optimal Aperture	2.17
Optimal Annulus	5.5
Transit Duration (day)	0.0802 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.226 $\pm$ 0.067 vs 0.16327 (deviation 0.82 σ)
Duration fit vs published	0.0802 d vs 0.0767 d (deviation 0.66 σ)
Mid-time O-C	0.9 min
Depth SNR	3.0
Residual scatter	72.56 %

Light curve

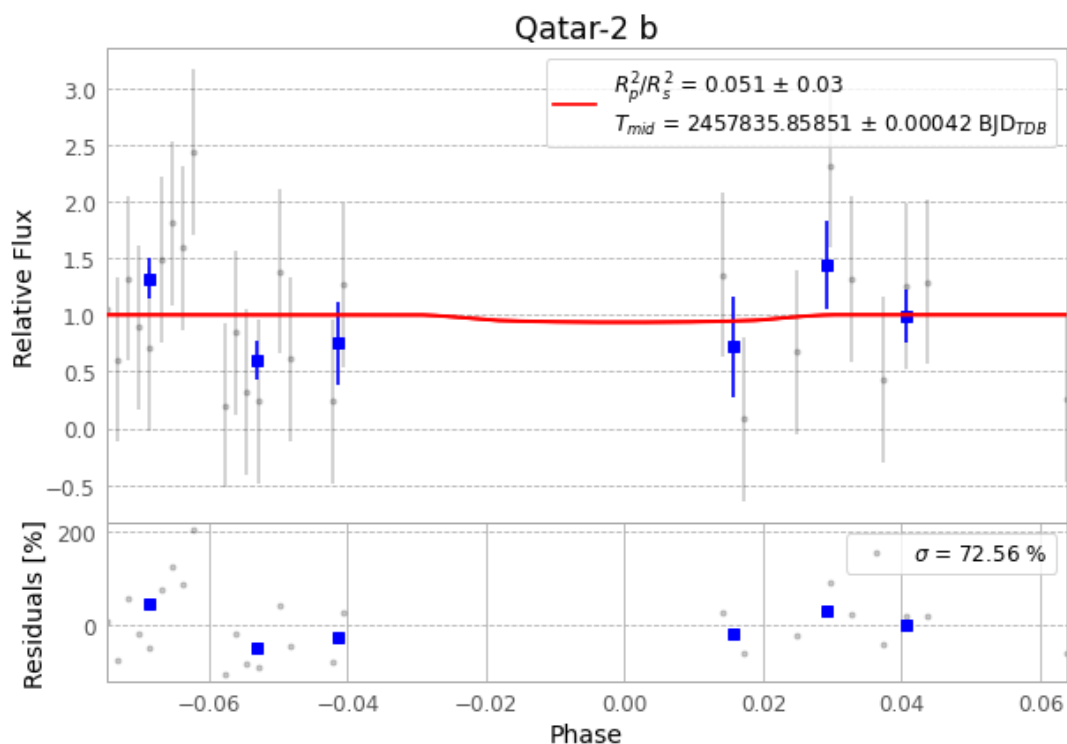


Figure 1 — FinalLightCurve_Qatar-2 b_2017-03-22.png (SHA-256 e1eab6b11717fd35...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [621, 219], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f9e20c4630549c49...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

99 FITS frames (65 MB), Qatar-2170323060313.FITS → Qatar-2170323110012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-170323.log (11eac6f4ef8b...), FinalLightCurve_Qatar-2 b_2017-03-22.png (e1eab6b11717...), FinalLightCurve_Qatar-2 b_2017-03-22.csv (0562d35216a0...)

Compute resources

Wall time 200.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 16:50Z.